张荐荣 Jianrong Zhang

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EDUCATION

State Key Laboratory of Integrated Chips and Systems, Fudan University **2025.09 - 2030.06**
Ph.D. Student majoring in EE, Supervisor: Prof. Liling Wang (Anticipated Graduation)

- Research Interests：Reconfigurable Computing、Computer Architecture
- ISCAS2026*1 (First Author)、DAC2025*1 (Second Author)、ISEDA2026*1 (Second Author, Best Paper Candidate)
- Homepage Link https://starryskyz.github.io

College of Integrated Circuits and Micro-Nano Electronics, Fudan University **2021.09 - 2025.06**
B.E. of Microelectronics

- GPA: 3.70/4.00 (Ranking: 20/124)
- Bachelor of Engineering with Honors (6/124)、Outstanding Graduate of Fudan University、Wangdao Scholar

School of Integrated Circuit, Peking University **2025.07**
Visiting Researcher, Supervisor: Prof. Yun (Eric) Liang

Nanyang Technological University, Singapore **2025.01**
Visiting Student

PROJECT EXPERIENCE

A Unified Full-Stack Framework for CGRA towards Model Inference **2025.08 - Now**
*ISCAS26(First Author) *1, another first-author paper is under review*

- A Plugin-friendly CGRA design, with a retargetable compiler backend towards different platform like Vitis, PYNQ or Cocotb.
- Further optimization on the MLIR frontend compiler, research about operator optimization strategies for ONNX model
- Seamless application-level CGRA verification, not only at the data level.
- End-to-end inference of an llama2-42M model on CGRA SoC with accelerating GEMV kernel, cut down the inference by 73.1 % than ARM A53 CPU, achieving 112MOPS/mw.

Optimization Library Design for X-EPIC Galaxsim SV RTL Simulator **2025.09 - 2025.11**
National First Prize (also Problem-05 Champion) of China Postgraduate IC Innovation EDA Elite Challenge Contest

- Analyzed bottlenecks in RTL simulation, like constant expression, redundant event scheduling, and complex signal hierarchy traversal.
- Developed multiple compiler opt techniques, including statement block merging, bit-width packing, alias folding, timing simplification.
- Leveraged advanced programming techniques such as functional programming, symbolic abstraction, and hash-based clustering and matching to improve compilation efficiency while significantly reducing runtime memory usage and execution time.
- Achieved over 15% performance improvement on industrial designs including the Xuantie C910 processor, substantially outperforming the runner-up in the overall competition evaluation.

AI-Enhanced Bayesian Optimization for Microarchitecture Exploration **2024.04 - 2025.09**
*Fudan Undergraduate Research Opportunities Program; DAC25 (Second Author) *1*

- Program-aware initialization for warm-starting exploration by leveraging prior knowledge of LLM in RISC-V and LLVM feature analysis.
- A novel multi-objective optimization that leverages an LLM as a surrogate model and incorporates an insightful sampler.
- A 22.8% improvement in energy efficiency with the same number of iterations and a 2.9× runtime speedup for the same target.

Implementation of Heterogeneous SoC of RISC-V & ARM **2023.06 - 2023.08**
2023 Fudan Micro Cup Undergraduate Electronic Design Contest, Team Leader

- Worked in a team of 3 to explore the collaboration between Programmable Logic and Processing System on ZYNQ.
- Implemented Rocket-chip on PL by using Chisel Rocket-chip SoC generator.
- Modified a Frontend Server and managed the exchange of data flow between PL and PS successfully.
- Accomplished the booting of Linux OS on both PL and PS, FPGA Overlayed.

PUBLICATIONS

- **An MLIR-Based Compilation Flow for CGRAs: Optimized CDFG Generation, Control-Flow Handling, and Scalability.**
  Jiahang Lou, Jiayao Hu*, **Jianrong Zhang**, Yikuan Chen, Zewei Zhong, Yuan Dai, Wenbo Yin, Lingli Wang#, Yu He, Qing He. International Symposium of EDA (ISEDA 2026), Singapore, 2026. Best Paper Candidate.*
- **Live Demonstration: an Agile FPGA-Overlayed CGRA SoC for High-Efficiency Computing.**
  Jiahang Lou, **Jianrong Zhang***, Yuan Dai, Zewei Zhong, Huan Lin, Wenbo Yin, Lingli Wang#. 2026 IEEE International Symposium on Circuits and Systems (ISCAS), Shanghai, China, 2026*
- **LEMOE: LLM-Enhanced Multi-Objective Bayesian Optimization for Microarchitecture Exploration.**
  *Jingyuan Li, **Jianrong Zhang**, Ye Li, Wenbo Yin, Lingli Wang#. 2025 62nd ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, USA, 2025.*

SKILLS

Language: Mandarin Chinese (Native); English (Fluent, CET-6: 593/710, CET-4: 637/710)
Programming Skills: C/C++, Python, Scala
Related Tools： Design Compiler; VCS; Vivado; Vitis; QT; Wireshark; Git

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教育背景

复旦大学 集成芯片与系统全国重点实验室 **2025.09 - 2030.06**

集成电路与系统设计方向博士研究生，导师：王伶俐 教授 (预计毕业时间)

- 研究方向：可重构计算及其相关软件方法、计算机体系结构
- 发表 ISCAS2026 会议论文 1 篇(第一作者)、DAC2025 会议论文 1 篇(CCF-A,第二作者)、ISEDA2026 论文 1 篇(第二作者 BP 候选)
- 个人主页 https://starryskyz.github.io

复旦大学 集成电路与微纳电子创新学院 **2021.09 - 2025.06**

微电子科学与工程 工学学士学位

- 总 GPA: 3.70/4.00(专业排名: 20/124)
- 复旦大学工学荣誉学士学位获得者(6/124)、复旦大学本科优秀毕业生、复旦大学望道学者

北京大学 集成电路学院 **2025.07**

访问学生，导师：梁云 教授

新加坡南洋理工大学 **2025.01**

交流学生，复旦 • 荣昶创芯之星 AI+集成电路科创训练营院系国际交流项目

项目经历

面向模型推理的统一全栈 CGRA 框架及自动化流程 **2025.08 至今**

ISCAS26 一作*1，另有一篇一作在投

- 实现 Plugin 的 CGRA 加速核，便于嵌入各类 SoC，并适配了一个统一的编译后端，可迁移生成不同平台的调用函数
- 进一步完善优化 MLIR 编译前端的数据流优化手段，和面向 ONNX 模型的算子调研优化策略
- 实现应用级无缝 CGRA 仿真验证，不止是数据层面的验证
- 实现 llama2-42M 模型的端到端运行，加速其中的 GEMV，单次推理延迟较 ARM A53 CPU 降低 73.1%，系统能效 112MOPS/mw

芯华章 Galaxsim SV 仿真器优化库设计 **2025.09 - 2025.11**

2025 中国研究生创“芯”大赛·EDA 精英挑战赛全国总决赛一等奖，赛题冠军

- 分析数字电路仿真中常见的常量计算赋值、事件调度冗余、信号层级复杂等瓶颈部分
- 开发了语句块合并、位宽打包、别名折叠、时序简化等多种优化手段
- 结合函数式编程、符号化抽象、哈希聚类匹配等高级编程技术，进一步提高了编译效率，同时显著降低运行时内存与用时开销
- 经验证在 Xuantie C910 处理器等工业级复杂案例中均取得 15%以上性能提升，综合评分大幅领先于第二名

AI 赋能的贝叶斯优化在处理器设计空间探索的研究 **2024.04 - 2025.09**

复旦大学本科生学术资助计划项目；DAC25 二作*1

- 利用大语言模型，实现程序特征感知和先验知识注入的热启动
- 以大模型作为含有先验知识的采样器和代理模型，进行更有洞察力的贝叶斯优化
- 在相同的迭代次数下，搜索效果提升了 22.8%；在实现相同目标的情况下，搜索速度提升了 2.9 倍

RISC-V 异构 SoC 架构设计与实现 **2023.06 - 2023.08**

2023 复微杯全国大学生电子设计竞赛 | 队长

- 以 3 人小队的形式探索了 ZYNQ 芯片 PL 和 PS 端的软硬件协作、数据交互、外设访问
- 在 PL 端利用 Chisel Rocket-chip SoC generator 实现了 RISC-V Rocket 核的部署
- 移植了一个前端服务器，并成功处理了 PL 和 PS 端的数据流交互及协议接口设计
- 最终在 PL 端和 PS 端都成功启动了操作系统，实现了一个 FPGA 上的异构 CPU SoC

学术论文

- An MLIR-Based Compilation Flow for CGRAs: Optimized CDFG Generation, Control-Flow Handling, and Scalability. Jiahang Lou*, Jiayao Hu*, **Jianrong Zhang**, Yikuan Chen, Zewei Zhong, Yuan Dai, Wenbo Yin, Lingli Wang#, Yu He, Qing He. International Symposium of EDA (ISEDA 2026), Singapore, 2026. Best Paper Candidate.
- Live Demonstration: an Agile FPGA-Overlayed CGRA SoC for High-Efficiency Computing. Jiahang Lou*, **Jianrong Zhang***, Yuan Dai, Zewei Zhong, Huan Lin, Wenbo Yin, Lingli Wang#. 2026 IEEE International Symposium on Circuits and Systems (ISCAS), Shanghai, China, 2026
- LEMOE: LLM-Enhanced Multi-Objective Bayesian Optimization for Microarchitecture Exploration. Jingyuan Li, **Jianrong Zhang**, Ye Li, Wenbo Yin, Lingli Wang#. 2025 62nd ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, USA, 2025.

兴趣与技能

语言： 普通话；英语 (流利, CET-6: 593/710, CET-4: 637/710)

编程技能： C/C++, Python, Scala 编程

专业工具： Design Compiler; VCS; Vivado; Altera; Gowin; QT; Wireshark; Git